

KHIZAR ANJUM
Machine Learning Engineer

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PROFESSIONAL EXPERIENCE

Graduate Research Assistant, CPS Lab — (2019 – Present) | Piscataway, NJ, US

- Specialized in developing efficient deep learning architectures for various applications, with focus on resource-constrained devices. Published 4 journal articles and 14 conference papers in prestigious venues including IEEE JBHI, IEEE JSAC and conferences like IEEE PerCom, MASS, and UComms.
- Filed 3 patents based on technologies developed the tenure.
- Helped secure funding worth \$1.135 million in prestigious NSF awards.

Research Assistant, Communications Lab — (2017 – 2019) | Lahore, Pakistan

- Developed and trained cutting-edge Convolutional Neural Network classifier with Bi-LSTM evidence aggregation model for Parkinson's detection on multimodal data.
- Dealt with diverse data-types such as gyroscopic measurements, voice recordings, tapping intervals, and memory test results that were highly imbalanced (Healthy vs. Parkinson's).
- Achieved 89% Area under ROC Curve performance compared to reference work performance of 56%, a 23% improvement.

RESEARCH INTERESTS

Analog and Neuromorphic Computing, Ultra-Low Power Neural Networks, Hardware-Software Co-design, FPGA Acceleration, Embedded Machine Learning, Computer Vision, Signal Processing, Underwater Acoustic Communications, Autonomous UAVs

PATENTS

- WO 2025/064997:** Anisotropic Diffusion-based Analog Neural Network Architecture
- WO 2024/107672:** Techniques for Image Transmission through Acoustic Channels in Underwater Environments
- (Pending Application):** Methods and Systems for Determining Group Motion Patterns

SELECTED RESEARCH PROJECTS

Ultra-Low Power Analog Neural Network Design for Health Monitoring — 2022-25

- Helped develop a fully-connected analog neural network design based on 3T1C design with a voltage output, dubbed Voltage-Resistive Processing Unit (VRPU), facilitating analog addition and demonstrating accuracy upto 96.57% on MNIST.
- Stacked existing VRPU design horizontally and vertically to form a functioning neural network by writing a custom Python-based generation script for ngspice circuit simulator, scaling complex hand-drawn circuits into programmable schematics.
- Developed the idea of Folded Neural Network (FNN), thereby trading off on-chip space and power-consumption for duty-cycle and demonstrating that a twice-folded 6-layer network with a hidden-size of 30 can be operated at an average power consumption of 53μW, thereby enabling batteryless operation.
- Invented a novel analog architecture called CvPU (Convolutional Processing Unit) based on anisotropic diffusion that can implement a convolutional layer with a 8-DoF 3×3 kernel when tessellated.
- Scaled the CvPU-based analog neural network up to 300×19 array sizes and estimated the power consumption to be around 55.58 mW with space consumption in the order of mm^2 .
- Trained the CvPU-based analog neural network EUH EEG Seizure Dataset demonstrating 78.5% detection accuracy.
- Tested and implemented the digital-side neural networks on FPGAs such as TinyFPGA BX and Xilinx Ultra96-v2 by writing custom neural computation modules using Chisel hardware construction language.

Crowd Prediction and Behavior Assessment Using Adaptive UAVs — 2024-25

- Leveraged efficient signal processing techniques by piggy-backing inter-prediction results from MPEG-4 hybrid video-encoder for crowd pattern identification to deliver 45× reduction in execution time as compared to baseline methods.
- Delivered comparable performance to Deep Learning and other optical flow-based baselines for crowd pattern identification task in terms of Average Angular Error (AAE), i.e., 14.99° on TUB CrowdFlow Dataset video sequences.
- Implemented automated exponentially converging implementation of Volkov Block Method for solving Laplace Equation, that can identify crowd hotspots/behavior based on domain knowledge. <https://github.com/khizar-anjum/pyBlockMethod>

Deep Joint Source Channel Coding for Underwater Image & Video Transmission — 2022-25

- Developed an LSTM-based sequence-to-sequence model to enable channel-state aware deep joint-source channel coding for transmitting images through underwater acoustic channel.

- Implemented a variable encoder that outperforms traditional JPEG/JPEG2000 approaches for compression in low-SNR regimes, i.e., from 0 to 13 dB, and avoids cliff-effects routinely observed in brittle traditional techniques.
- Performed real-life experiments at Rutgers Olympic pool for transmitting underwater images.
- Did a live demonstration at ACM WUWNet 2022 in Boston, MA showing live image transmission.
- Collected first-ever real-life acoustic communication dataset under various center frequency, bandwidth, and TX/RX configurations at Barnaget Bay, NJ and Atlantic City, NJ and published the dataset as ACommSet.
- Developed a video multicast modulation-free approach based on end-to-end training of an auto-encoder architecture that learns the waveforms directly and eliminates 12.9% metadata overhead.

On-board Inference on Unmanned Aerial Vehicles — 2020-25

- Developed and implemented Convolutional Bi-LSTM auto-encoder for fault-cause detection on a PYNQ-Z1 FPGA yielding 40× speedup on-board (runtime of 2.63 msec) with a 96.25% test accuracy for real-life propeller-fault experiments.
- Designed a multi-UAV situational awareness algorithm for real-time detection of a suspect/weapon and tested using Microsoft AirSim simulator via hardware-in-the-loop emulations under varying network conditions.
- Published a novel context-aware framework for efficient and robust segmentation on-board demonstrating an mIOU of 0.95 on varied simulated weather conditions such as sunny, foggy, and rainy.
- Developed a real-time video-based navigation framework for drones using adaptive parameter selection for traditional models as well as neural networks, demonstrating 20-70% reduction in execution time with only 2% of accuracy drop.

SELECTED PERSONAL PROJECTS

Communication-aware Large Language Model Training on HPC Cluster — 2025

- Developed passive and active communication profiling techniques for training large language.
- Trained and profiled llama-3-8B and llama-3.3-70B-Instruct models using Megatron-LM on NERSC Perlmutter GPU cluster.
- Observed communication patterns for pipeline-parallelism, model-parallelism, tensor-parallelism, and data-parallelism.

Model Context Protocol (MCP) Design for Security & Legal Applications — 2025

- Won 3rd prize in an open-hackathon for designing an MCP server for cybersecurity vulnerability assessment, providing tools to search GitHub repositories, query NIST NVD, access CISA's Known Exploited Vulnerabilities catalog, and analyze repository contents for CVE research and proof-of-concept discovery. <https://github.com/khizar-anjum/risky-business-mcp>
- Designed a powerful MCP server for legal research across 3,352 U.S. courts using the CourtListener API, enabling access to millions of legal documents, court opinions and PACER data. <https://github.com/khizar-anjum/courtlistener-mcp>

EDUCATION

Rutgers University, New Brunswick, NJ, (2019–2026 [Expected])

- PhD Candidate & Master of Science (MS), Electrical and Computer Engineering (ECE)
- Grade Point Average (GPA): 3.92/4; *Advised by* Dr. Dario Pompili

Lahore University of Management Sciences (LUMS), Lahore, Pakistan (2015–2019)

- Bachelor of Science (BS), Electrical Engineering (EE) – Gold Medalist
- Grade Point Average (GPA): 3.86/4; *Advised by* Dr. Muhammad Tahir and Dr. Momin Uppal

RELEVANT SKILLS

- **Programming Languages:** Python (advanced), C++, Chisel HDL, MATLAB, Bash, SPICE (ngspice)
- **ML & AI Tools:** PyTorch, TensorFlow, Megatron-LM, Scikit-learn, OpenCV, Pandas
- **Hardware & FPGA:** Hardware & FPGA: Xilinx Vivado, PYNQ, TinyFPGA, Chisel language, FPGA deployment
- **Edge Computing & DSP:** GNU Radio, ROS, AirSim, low-power analog-digital ML systems
- **HPC & Distributed Computing:** SLURM, NERSC Perlmutter, distributed training (pipeline/tensor/data parallelism)
- **Development & Deployment:** Docker, Git/GitHub, Linux, Visual Studio Code, Jupyter, Model Context Protocol (MCP)

ACHIEVEMENTS & AWARDS

- Recipient of **Rutgers Innovation Awards** in the Graduate Student Category, 2025
- Recipient of **Paul Panayotatos Scholarship** for Sustainability-aware Electrical Engineering, 2024
- Selected for **NSF NOVUS & NSF I-Corps** program for innovation & customer discovery training, 2022
- Awarded **Travel Grants** from Rutgers University and ACM for **ACM WUWNet** and **IEEE MASS** participation
- Recognized with **Gold Medal** at LUMS for academic excellence, 2019
- Win: **TA Achievement Award**, 2020 from Rutgers